Lesson 3: Keep the player on screen

Pygame CE • S3 onwards • 40-50 min

Learning intention
Use comparisons and min/max to enforce screen boundaries.

Success criteria

- I can stop the shape leaving the window.
- I can explain why the limit is `WIDTH - SIZE`.

Key ideas

- A shape's x and y give its **top-left corner**, so the right edge is x + SIZE.
- `max(0, ...)` sets a lower limit; `min(..., limit)` sets an upper limit.

Activities

1. Move the square off screen and discuss which values cause this.
2. Add the boundary rule after movement is calculated and before drawing.
3. Test all four edges and all four corners.

Code

Key change

```
x = max(0, min(x, WIDTH - SIZE))
y = max(0, min(y, HEIGHT - SIZE))
```

Complete program

```
import pygame

pygame.init()

WIDTH = 800
HEIGHT = 600
SIZE = 40

screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("Stay on screen")
clock = pygame.time.Clock()

x = 380.0
y = 280.0
speed = 250
running = True

while running:
    dt = clock.tick(60) / 1000

    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    x += (keys[pygame.K_RIGHT] - keys[pygame.K_LEFT]) * speed * dt
    y += (keys[pygame.K_DOWN] - keys[pygame.K_UP]) * speed * dt

    x = max(0, min(x, WIDTH - SIZE))
    y = max(0, min(y, HEIGHT - SIZE))

    screen.fill((30, 30, 40))
```

```
        pygame.draw.rect(screen, (80, 200, 120), (int(x), int(y), SIZE, SIZE))
        pygame.display.flip()

pygame.quit()
```

Checkpoint

The square stays fully visible at every edge.

Support and stretch

- **Support:** Do the left edge together, then learners complete the other three edges.
- **Stretch:** Add a smaller play area inside the window, or an obstacle wall.

Reflect

Explain in your own words why `WIDTH` would be wrong as the right limit.

Save your work
Save this version as `lesson3.py` before you start the next lesson.